

Project Interim Report
On
E-Commerce Shopping Cart System
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming With Java

SESSION: 2025-26



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Abstract :-

The E-Commerce Shopping Cart System is a Java-based application developed using Object-Oriented Programming concepts. This project allows users to add and remove products, view items in the cart, and generate a final bill.

The system is designed to simplify shopping calculations by automatically computing total price, applying discounts, and adding taxes. It reduces manual effort and ensures accurate billing.

This project demonstrates real-world implementation of OOP concepts such as classes, objects, and encapsulation while handling product data efficiently.

Introduction:-

E-commerce platforms require a shopping cart system to manage products selected by users. This project simulates a simple shopping cart where users can add or remove products and view their cart.

The system calculates the total cost of items, applies discount and tax, and generates a final bill. It helps in understanding how real-world shopping systems work and demonstrates the use of Java programming and OOP concepts.

Scope:-

This project is suitable for small-scale shopping systems. It can be extended by adding database support, graphical user interface, and user login features in the future.

System Requirements:-

Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus)

Methodology:-

First, the user adds products to the cart by entering product details such as ID, name, price, and quantity.

The system stores the products in a list. Users can also remove products using their ID.

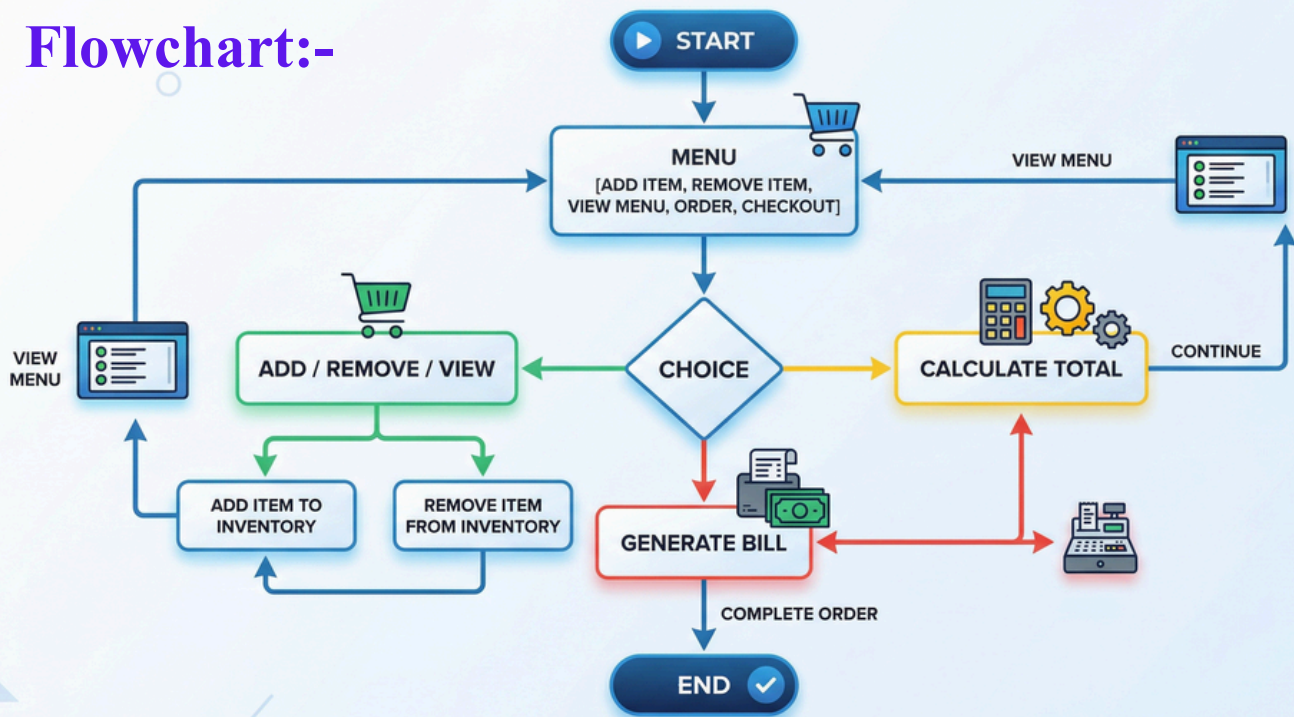
The total price is calculated by summing the price and quantity of all products. A discount of 10% is applied, followed by 5% tax.

Finally, the system displays the complete bill including total amount, discount, tax, and final payable amount.

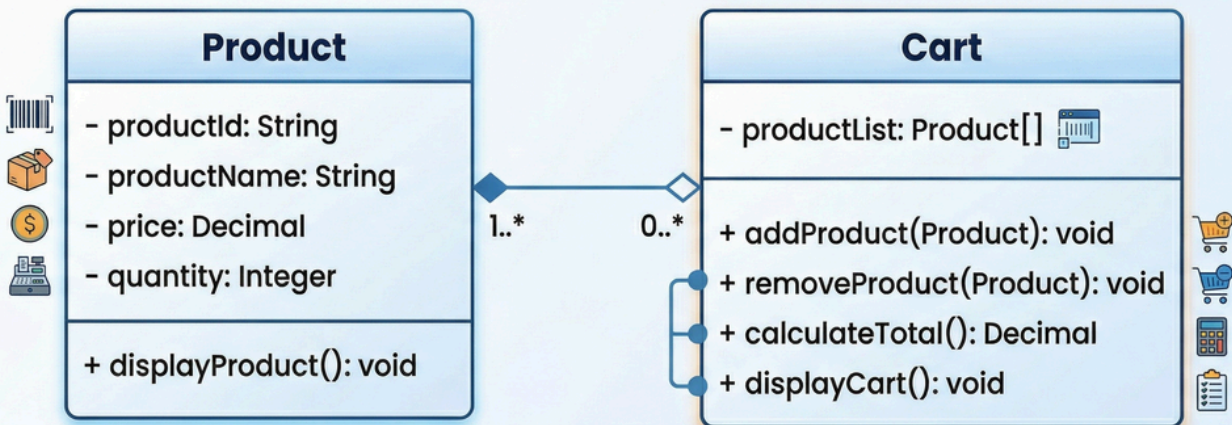
Objective:-

- To add and remove products from cart
- To calculate total price of products
- To apply discount and tax
- To generate final bill
- To implement Object-Oriented Programming concepts

Flowchart:-



SYSTEM CLASS DIAGRAM: PRODUCT & CART



Start → Menu → Add / Remove / View → Calculate → Bill → End

Product Class:

- productId
- productName
- price
- quantity
- displayProduct()

Cart Class:

- product list
- addProduct()
- removeProduct()
- calculateTotal()
- displayCart()

Implementation:-

The project is implemented using Java and follows OOP concepts.

The Product class stores details of each product including name, price, and quantity.

The Cart class manages operations like adding products, removing products, displaying cart items, and calculating total price.

The main class handles user interaction and performs billing operations including discount and tax calculation.

Future Enhancement:-

- Add graphical user interface (GUI)
- Implement database storage
- Add user login system
- Add coupon and discount codes

Sample Input / Output:-

```
File Edit Selection View Go Run ... Java project

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\Java project> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.10.7-hotspot\bin\java.exe'

1. Add Product
2. Remove Product
3. View Cart
4. Checkout
5. Exit
Enter choice: 4

Total: ?2000.0
Discount: ?200.0
Tax: ?90.0
Final Amount: ?1890.0

1. Add Product
2. Remove Product
3. View Cart
4. Checkout
5. Exit
Enter choice: 2
Enter Product ID to remove: 1
Product removed!

1. Add Product
2. Remove Product
3. View Cart
4. Checkout
5. Exit
Enter choice: 5
Thank you!

PS E:\Java project>
```

```
File Edit Selection View Go Run ... Java project

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\Java project> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.10.7-hotspot\bin\java.exe'

1. Add Product
2. Remove Product
3. View Cart
4. Checkout
5. Exit
Enter choice: 1
Enter ID: 1
Enter Name: Shirt
Enter Price: 1000
Enter Quantity: 2
Product added!

1. Add Product
2. Remove Product
3. View Cart
4. Checkout
5. Exit
Enter choice: 3
Shirt - ?1000.0 x 2 = ?2000.0
```


Gantt Chart:-

Task		Time
Planning		Day 1
Coding		Day 2-3
Testing/Debugging		Day 4
Report		Day 5

Conclusion:-

The project successfully demonstrates the use of Java and OOP concepts. It provides a simple and efficient shopping cart system that performs accurate billing operations.

References:-

- [1] H. Schildt, Java: The Complete Reference
- [2] Oracle Java Documentation
- [3] www.google.com
- [4] www.geeksforgeeks.com